Alexander Yao

(908) 381-9450 | alex.r.yao@gmail.com | linkedin.com/in/alexyao2 | github.com/alexyao2 | alexyao2.github.io/portfolio

Education

University of Maryland College Park, MD

Bachelor of Science in Computer Science, Minor in Applied Mathematics *August 2023 – Expected December 2026*

- **Relevant Coursework**: Probability & Statistics, Algorithms, Advanced Data Structures, Linear Optimization

Experience

Data Science Analyst August 2026 – Present

EMI Advisors *College Park, MD*

- Assisting with developing dashboards, reports, and data visualizations for client deliverables.
- Contributing to machine learning and predictive analytics projects under the guidance of senior team members.
- Designed a Visio flowchart mapping key project aspects including data preparation, model evaluation, and proposed deployment; presented it to supervisors and senior team members for use in a client meeting.

Data Analyst Intern May 2026 – August 2026

Icarus Fund, LLC *New York, NY*

- Built Python pipelines integrating options and market data, processing over 146M matched records and comparing recalculated option risk measures against vendor values across 5M+ rows.
- Developed an options backtesting pipeline to evaluate trading signals, account for transaction costs, and restrict signal inputs to data available at the decision time.
- Assessed strategy robustness using 200 randomized permutations and sensitivity analysis across three dealer-share assumptions; generated reproducible reports documenting performance and validation results.

Computer Network Research Assistant May 2025 – February 2026

George Mason University *Fairfax, VA*

- Built a Python client-server testbed for live webcam video and microphone audio streaming over QUIC using asynchronous programming.
- Implemented media capture, encoding, receiver-side buffering, and playback using OpenCV and Opus, with optional QUIC event logging for debugging.

Projects

NBA Referee Analytics | *AWS, React, Vite, FastAPI, Python, Docker* June 2026

- Built an interactive React and FastAPI application to compare NBA referee statistics across seasons, with filters for season, game category, and games officiated.
- Implemented game-weighted averages, foul differentials, z-score outlier detection, and cross-season variability comparisons, including a custom score combining home-team outcomes and foul patterns.
- Containerized the frontend and backend with Docker Compose and configured Nginx to route application and API requests.

Trade War Monitor | *Python, SQLite, FastAPI, React, TypeScript* September 2026

- Built an hourly Python pipeline to collect four market indicators spanning energy, freight, and gold, validating inputs and storing available observations when individual sources fail.
- Implemented rolling anomaly scores for prices and daily changes using historical baselines, with checks for insufficient history and outdated data.
- Developed an interactive dashboard linking sourced political events to historical Brent prices through synchronized maps and timelines, distinguishing price movements from cause and effect relationships.

Technical Skills

Programming: Python, SQL, JavaScript, TypeScript, Java, C
Data & Analytics: pandas, NumPy, PyArrow, DuckDB, SQLite, statistical analysis, data validation
Application Development: FastAPI, React, SQLAlchemy, REST APIs
Tools & Infrastructure: Docker, Docker Compose, Nginx, Git, Linux